



OPTICAL COMMUNICATIONS

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3 – The optical fiber

Structure, manufacturing, attenuation

Guided light

- 1854: John Tyndall presented to the Royal Society how **light could be guided** through the jets of water in fountains, thus demonstrating the possibility of **"bending" light.**



- 1961: Elias Snitzer theorized the functioning of an **optical fiber capable of transmitting light with minimal losses.**

- 1964: Charles K. Kao (University of New South Wales, Australia) understood the need to **eliminate impurities from glass** to build optical fibers for TLC.



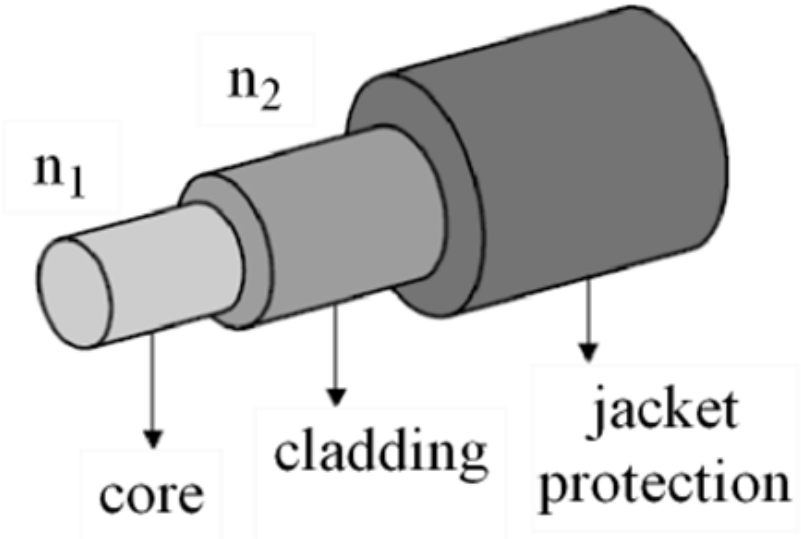


- In 1970 R. Mauer, D. Keck and P. Schultz (Corning Glass) produced an optical fiber made of **extremely pure silicate glass with low losses**

Optical fiber structure

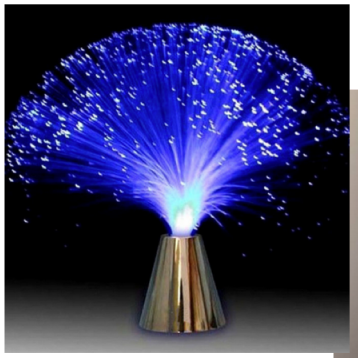
An optical fiber consists of a **thin cylindrical strand** of **core** with refractive index n_1 surrounded by a **cladding** whose refractive index n_2 is slightly lower than that of the core.

The specific structure of the fiber allows to guide the ligth inside the core.



In case of water jets, the cladding at lower *n* is contituted by the surrounding air.

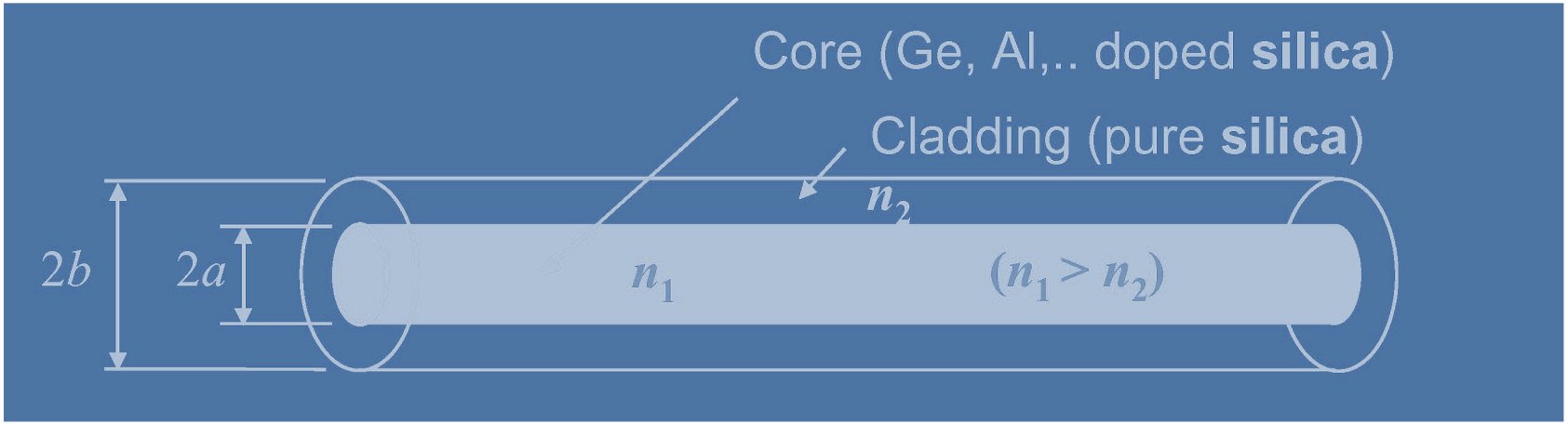
Optical fibers in plastic find also applications for furnishing and entertainment uses.





Optical fiber in glass for TLC

The optical fiber for telecommunications is a dielectric waveguide realized in **fused Silica SiO_2**



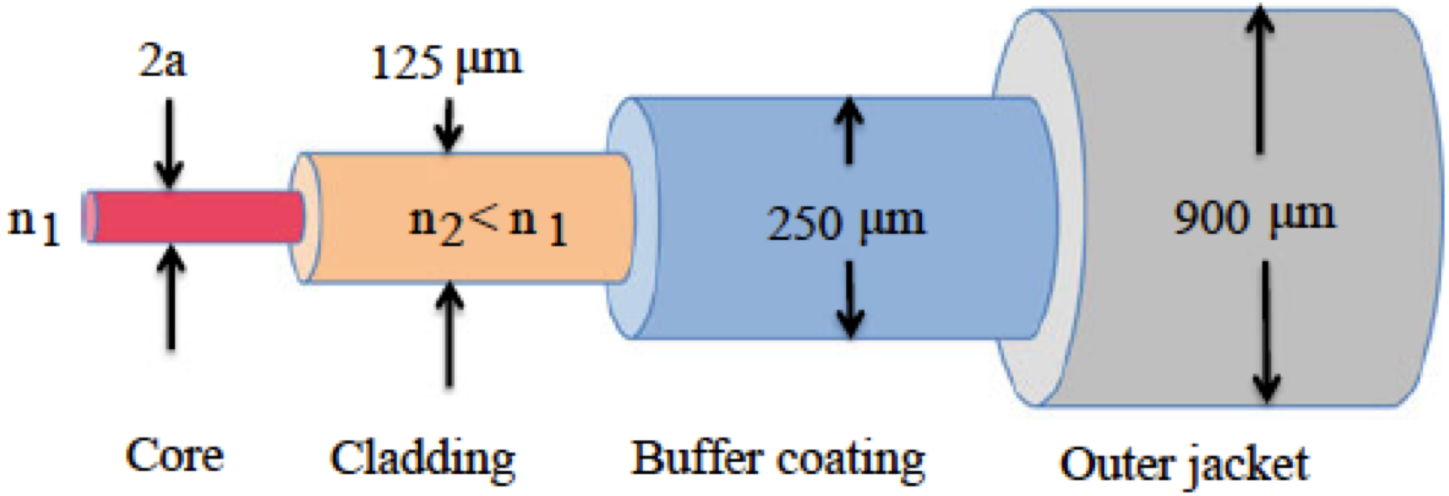
CORE: realized with **SiO_2** with doping materials (Ge) -> typical n_1 = 1.458
CLADDING: realized with only **SiO_2** -> typical n_2 = 1.448
Typical index step: $\Delta n \approx 0.001 - 0.01$

Optical fiber in glass for TLC

Typical dimensions:

CLADDING DIAMETER: 125 μm
CORE DIAMETER: 9-10 μm for single-mode fibers
: **50-62.5 μm** for multi-mode fibers

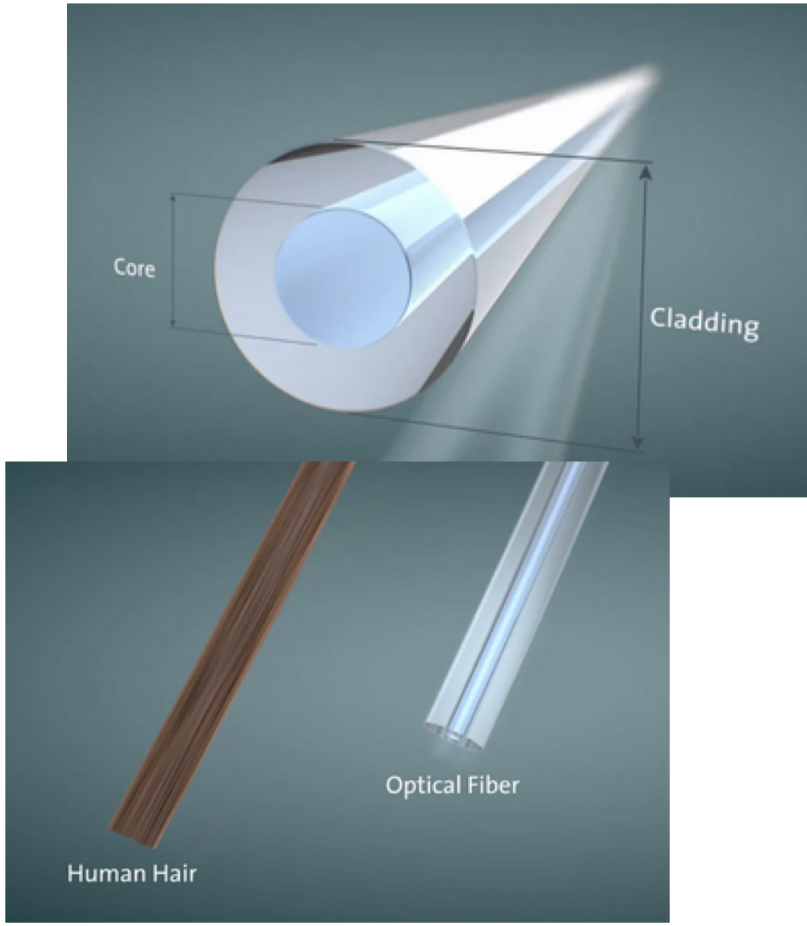
OVERCOATING DIAMETER: 250 μm



Optical fiber in glass for TLC

BENEFITS

- Small size and low weigth
- Flexibility
- Strong tensile strength
- Immunity to electromagnetic interference
- Low cost

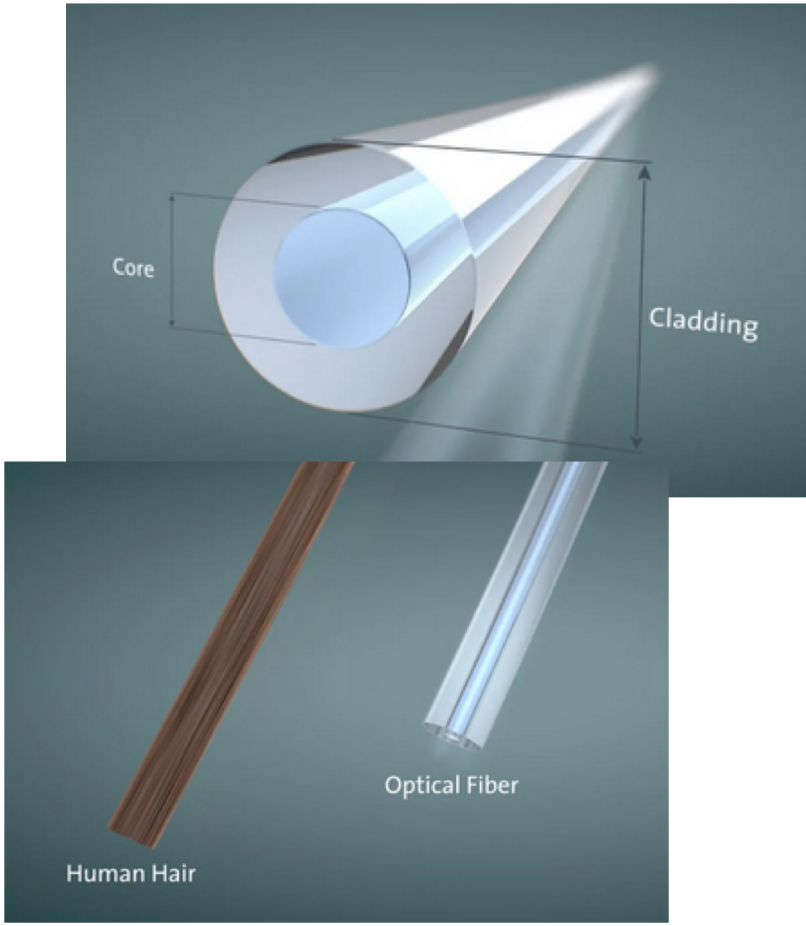


- Long distance transmission (very low attenuation)
- Large information capacity (transparent to the modulation rate)
- Increased signal security (quantum communications)

Optical fiber in glass for TLC

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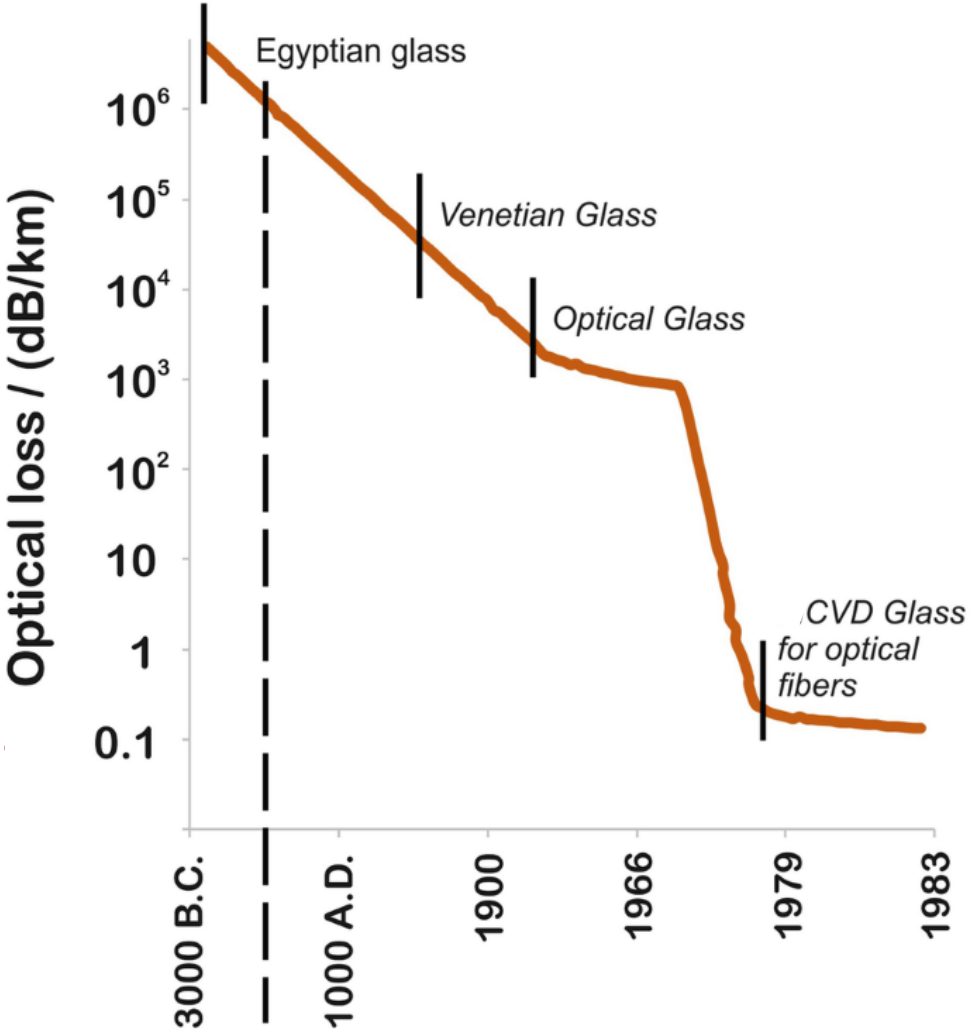


Why does the TLC fiber glass shows such low α?

The optical fiber glass is NOT obtained by fusion of the glass-bare, but by *chemical synthesis*.

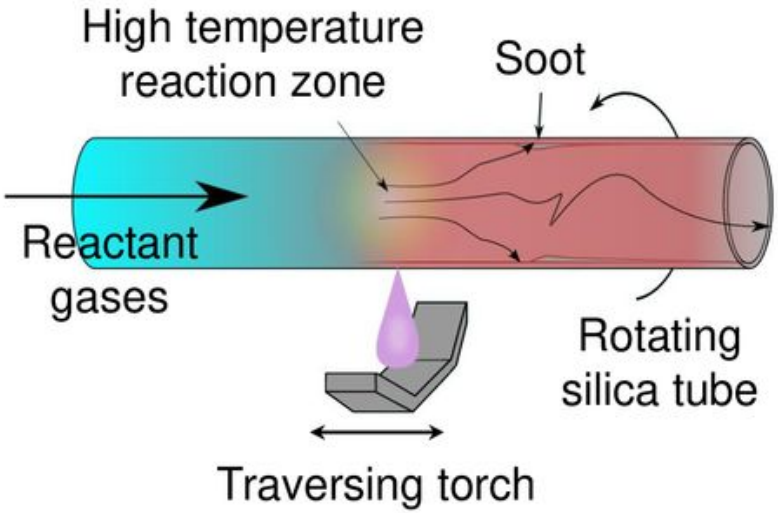
The chemical reaction occurs by burning SiCl4 in an atmosphere rich of oxygen.

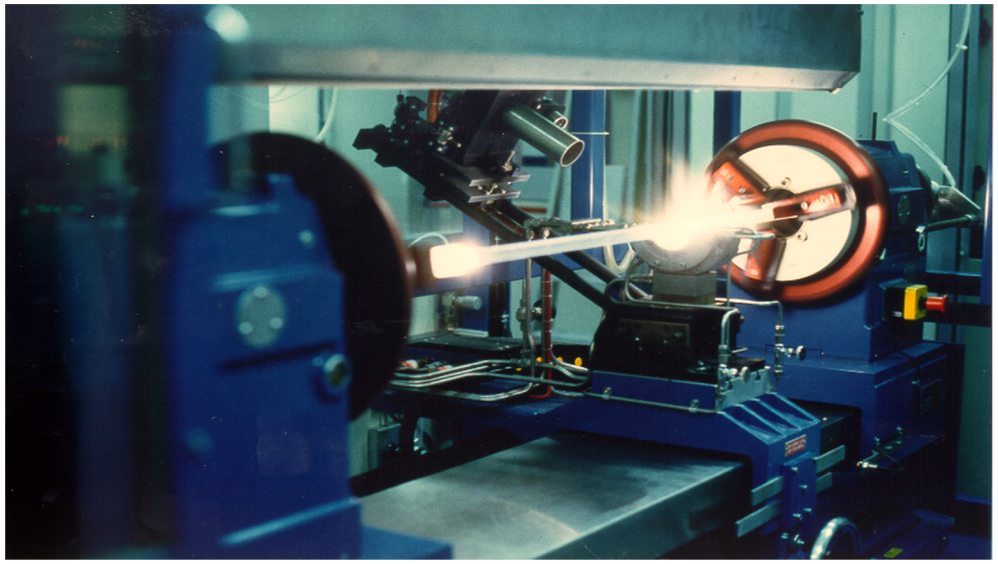
Since the chemical reactant gases are very pure, the realized glass is very very pure, *with very low attenuation α in dB/km.*



Chemical Vapor Deposion - CVD

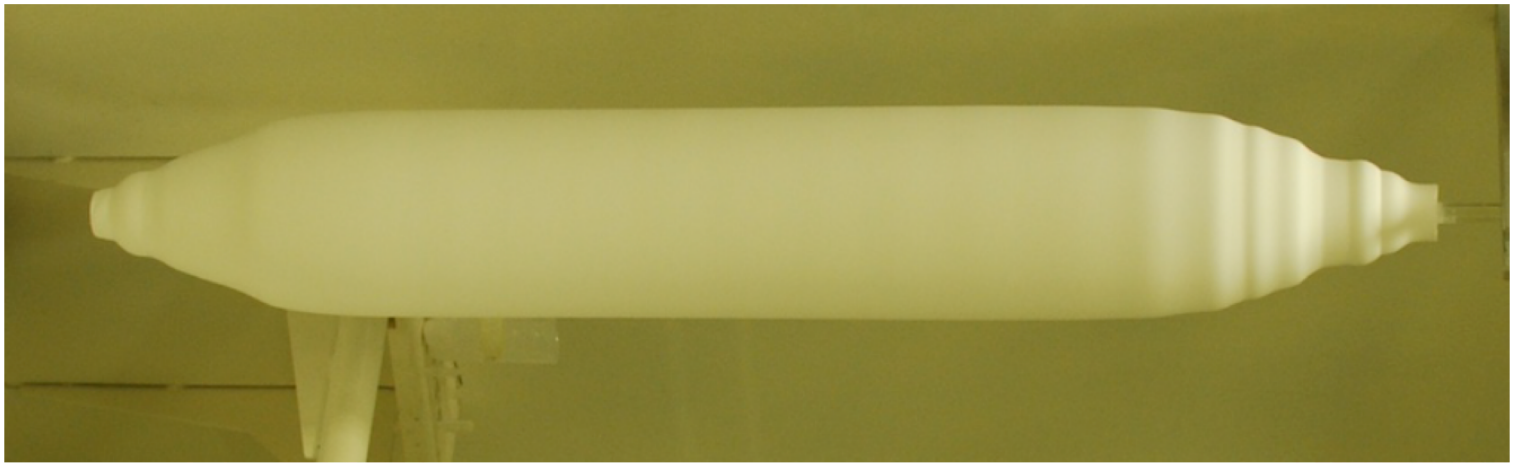
The process is known as **CHEMICAL VAPOR DEPOSITION (CVD)** and occurs in a rotating tube (after removed), stimulated by a torch. In the high temperature reacting zone, the inserted ultra-pure reactant gases produce the silica glass.





Chemical Vapor Deposion - CVD

After several passes of the tube over the torch, which brings it to temperatures above 2000°C, the produced glass collapses, closing the internal opening and degenerating into a full cylinder, called **PREFORM**.

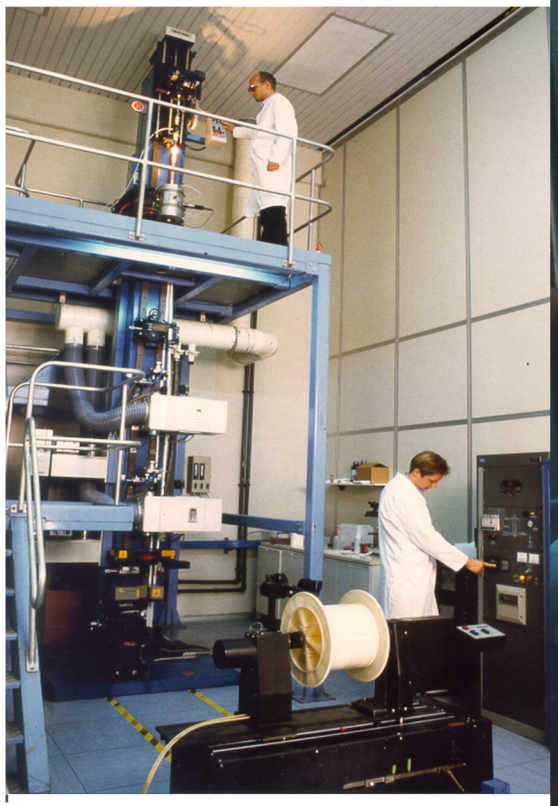


The preform internally already presents the innermost part with a higher refractive index n_1 and the external part with a lower index n_2, with diameter dimensions characterized by the *same proportions* that will be obtained in the final fiber.

Fiber spinning

To achieve the gradual reduction of the preform diameter (from around few cm) without altering the index profile and geometric ratios, the preform is inserted at low speed inside the furnace in a «spinning tower» and is heated in a small area to temperatures above 2000°C.

The dimensions and thermal profile of the heated zone are carefully optimized as they influence the mechanical, geometric and optical characteristics of the fiber.



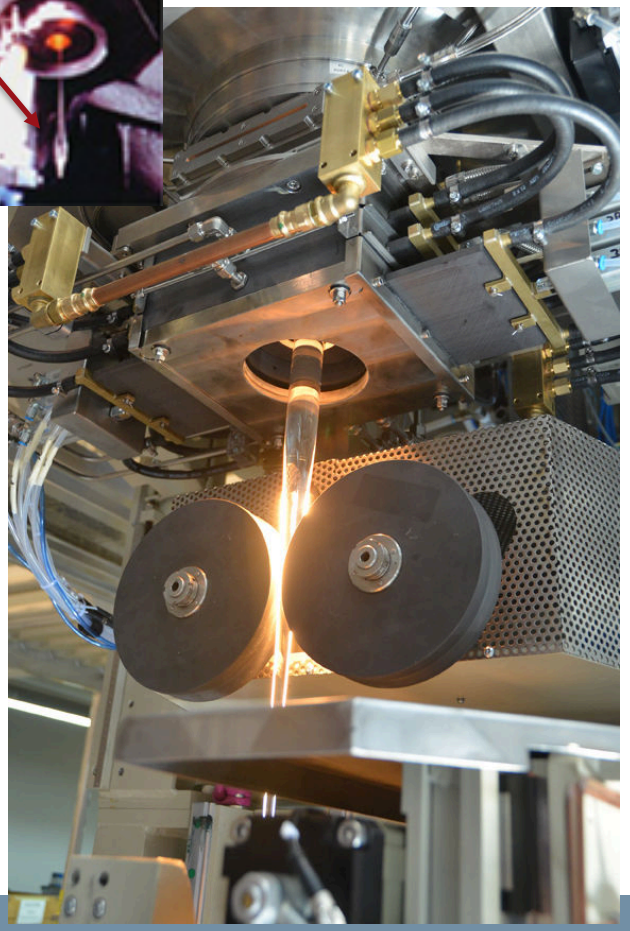


Fiber spinning

In the hot zone the preform reaches the 'softening' temperature. The «hot drop falls» at a speed higher than the insertion speed, causing the radial dimensions to decrease.

The «glass wire» goes down from the tower by gravity, gradually cooling and spinning the preform.





Fiber spinning

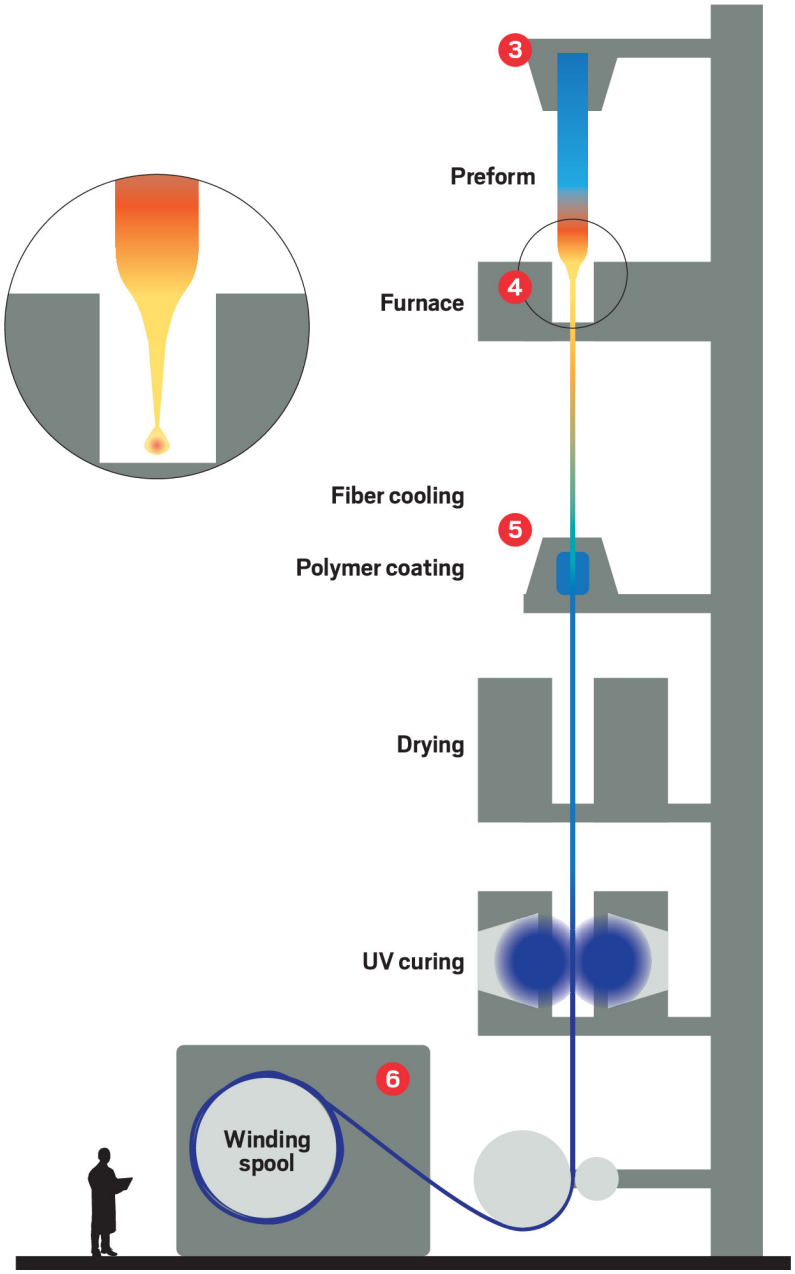
Once down, it has reached the target fiber size and can be wound into a suitable winding spool, after covering by a plastic coating (usually acrylate) to protect the fiber from mechanical breaks (the fiber without coating is very fragile).

For the conservation of diameters proportions:

$$\frac{d}{D} = \left(\frac{V}{\nu}\right)^{1/2}$$

D preform diameter
d fiber diameter
V preform insertion speed
ν fiber pulling speed

More than 10 km spinned from a preform rod

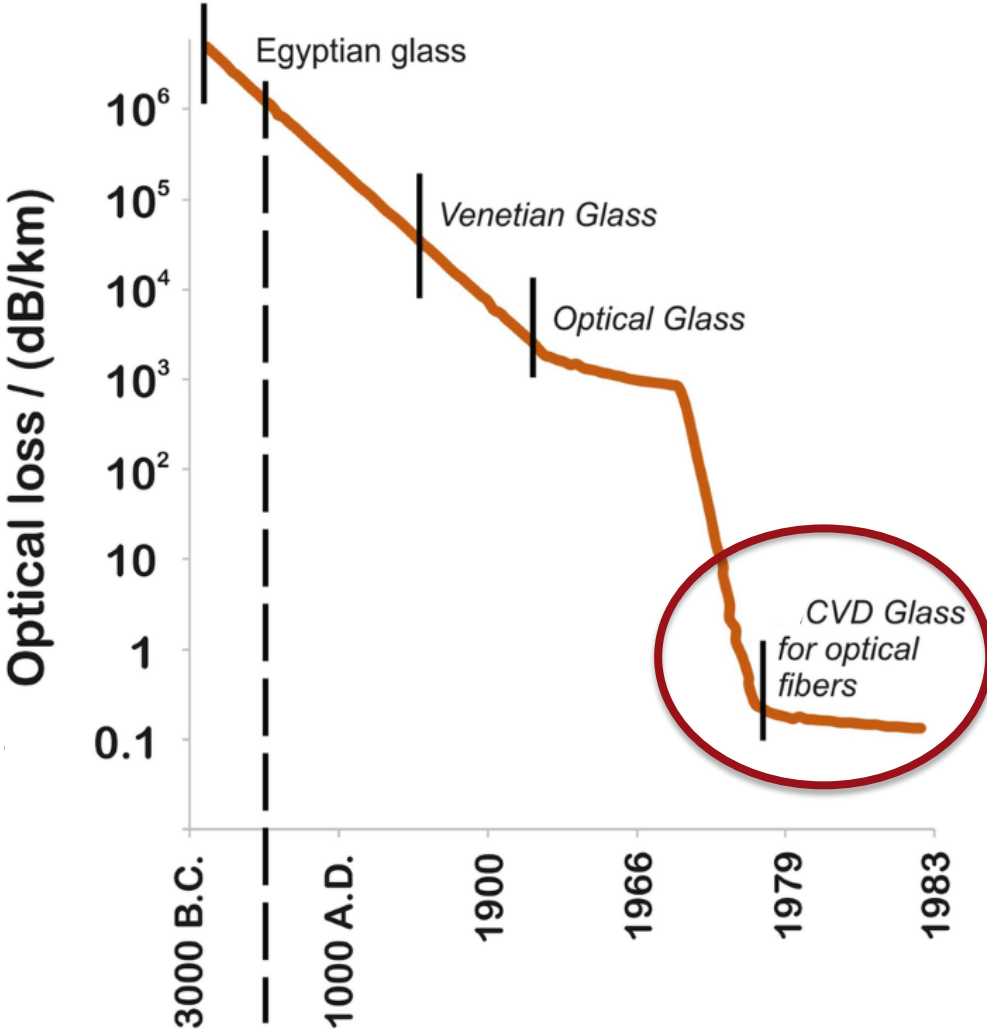


Why does the TLC fiber glass shows such low α?

The CVD process is the MAIN reason of the glass fiber success in TLC.

The MCVD glass is ultra-pure with a transmission window from 200 to 3500 nm.

Corning Glass Corp. in USA was the first company to optimize the process (many Patents)



ATTENUATION in optical fibers

The attenuation is the main issue of the optical fiber , but at the same time the reason of the optical fiber success in communication systems.

$$A_{dB} = 10log_{10}\left(\frac{P_{out}}{P_{in}}\right) = -\alpha_{dB} \cdot L$$



$$\alpha_{dB} \cdot L = 10log_{10}\left(\frac{P_{in}}{P_{out}}\right)$$

$$\alpha_{dB} = \frac{10}{L} \cdot log_{10}\left(\frac{P_{in}}{P_{out}}\right) \quad \text{in} \quad {}^{dB}/_{km}$$

Why the ATTENUATION is a critical issue?

In the *power budget* of the transmission system, the SENSITIVITY is defined as the power at the receiver assuring the target performance (BER).

$$L_{max} = \frac{10}{\alpha} 10 log_{10} \left(\frac{P_{in}}{P_{out}} \right)$$



The maximum length L_{max} of the transmission link depends linearly by the attenuation α and only in a logarithmic way by the input power P_{in}.

The gain in adopting low-attenuation fibers is larger than the gain in adopting higher input power.